# Endpoint Monitoring and Automated Containment Architecture for a Simulated Security Breach

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# Executive Summary

Organisations are increasingly relying on distributed computing devices e.g. workstations, servers and remote endpoints. Cyber attackers often target these endpoints since they saving user credentials and grant consumers direct access to corporate networks. Keeping watch on endpoint activity and responding promptly to abnormal behaviour are thus critical towards ensuring organisational security.

In this project, a controlled endpoint monitoring and automatic containment system was implemented on a simulated enterprise scenario. Larkspur Retail Group is a fictional organisation that has Windows and Linux systems and has recently shown signs of suspicious administrative actions. The purpose of this project was to reproduce this kind of activity in a controlled laboratory environment and introduce control and reaction systems that could identify and limit these menaces (M. Cinque, 2018).

A virtualised laboratory was configured that contained a Linux monitoring server and a windows endpoint. Wazuh agents gathered event data created by the endpoints and analysed it using the Wazuh SIEM platform. Detection logic has been created to detect privilege escalation and account manipulation behaviours.

Docker containers were utilized to create an automated remediation layer in order to improve response capabilities. The analysis of the alerts produced by the SIEM was performed by a lightweight Python-based AI component to decide whether remediation measures must be implemented. When a simulated attack with the creation of a new administrative account was conducted, the system accurately detected the activity and issued pre-defined remediation commands (M. A. Ayu, 2023).

The findings prove that monitoring via SIEM and an integration into containerised automation processes can meaningfully positively affect incident response time and the security team burden.

# 1. Introduction

## 1.1 Background

Enterprise networks are rife with endpoints that are dealing with sensitive data and organisational services. Endpoints are frequently a favourite target to attackers since, when hackers get user credentials or set improper permissions, they have a route to more privileged access.

Account manipulation and privilege escalation are especially harmful methods as they enable the attackers to gain administrative access to the systems. When privileged access has been gained then attackers may install persistence mechanisms, alter system configurations, or gain access to sensitive information (S. K, 2024).

Security observation tools like SIEM systems are significant in identifying these activities by consolidating records across several hosts as well as correlating activities within the environment. Singer and Friedman, (2013) assert that the centralised log analysis enhances better visibility of malicious behavior and facilitates quicker incident investigation.

Besides monitoring, automated response technologies are being increasingly implemented in organisations. Through automation, security teams are able to respond instantly to a threat without the need of manual intervention. Nevertheless, robot procedures should be carried out attentively to avoid unintentional interference of legitimate operations (M. Ç. Fanuscu, 2022).

## 1.2 Objectives and Scope

This project thus examined how a Security Information and Event Management (SIEM) platform with integrated automated containment logic could detect suspicious administrative activity and respond in a defensible, reversible way. The main goal was to build and evaluate an endpoint monitoring system that could detect privilege escalation and account manipulation activities on the simulated enterprise network. The project involved setting up endpoint telemetry collection, a SIEM-based monitor using the Wazuh platform and a remediation layer automation with services running in containers. The project was limited to a controlled lab set with one Windows endpoint and one Linux monitoring server, providing an isolated method where detection rules and automated response logic could be tested in a sandbox without causing disruption on real world infrastructure.

## 1.3 Out of scope

This project was also bounded in terms of several areas of enterprise security architecture. Network segmentation was neither redesigned nor were enterprise-wide endpoint detection and response systems put into effect, nor was a full forensic investigation performed on multiple hosts. It lacked the inclusion of email security controls, firewall rule optimisation, or industry scale threat intelligence integration. As a result, the attention remained solely on endpoint monitoring, SIEM-based analysis of dubious administrative actions as well as to create a lightweight automated remediation framework in the simulated laboratory.

## 1.4 Report structure

The report is structured as follows to present clearly its design, implementation and evaluation of the security monitoring system. Create a project introduction that includes researchers, objectives and scope. The literature review includes relevant cybersecurity concepts, including logging and detection engineering approaches, SIEM architectures, as well as automated response mechanisms. The scenario and threat model defines the imitation organisational context and synthetic attack conditions under which the experiment took place. The methodology section details the associated laboratory setup, telemetry sources and detection logic implemented in the SIEM environment. The implementation section describes the architecture of the monitoring and auto-remediation system. The results section discusses the outcome of both the simulated attack and detection process, whereas discussion critically evaluates how effective this system is. Lastly, the report ends with recommendations and a summary of project findings.

# 2. Literature review and standards alignment

## 2.1 Endpoint Security Principles

Endpoint security is a fundamental component of modern cybersecurity strategies because endpoints such as workstations, servers, and mobile devices represent common entry points for attackers. A good endpoint protection should be grounded in the multiple principles of least privilege, attack surface reduction, application control and secure logging. Following the principle of least privilege, users and processes have minimal permissions needed to carry out their functions limiting the potential damage done by compromised accounts. Through attack

surface reduction, the use of unnecessary services is disabled and system functionalities that could be abused are disallowed. Application control methods determine which specific software can run on endpoints, and block malicious or unwanted applications from running. Secure logging is equally important since it provides system activity visibility as well as enables security teams to identify abnormal behaviour. As stated by Singer and Friedman (2013), proper monitoring and logging practices can play an extremely important role in the early identification of malicious actions, thus allowing for timely incident response (T. Rahmawati, 2023).

## 2.2 Logging and Detection Engineering

Logging and detection engineering are the bread and butter for finding bad actors in enterprise systems. Learning by example, it is worth noting that Endpoint Detection and Response (EDR) tools are heavily dependent on numerous types of telemetry data collected from endpoints such as process creation events, authentication logs, file changes, network connections. Logging tools like Microsoft Sysmon extend native Windows logging functions, recording in-depth details of systems activity (process execution, registry modification). For Linux environments, monitoring solutions like auditd offer analogous functionalities that log every authentication request, command execution, or privilege escalation. You analyse this telemetry and design rules that can detect potentially suspicious patterns or behaviours. The bulk of detection strategies are organized by the MITRE ATT&CK framework, which tracks techniques and tactics as adversary methods observed in the wild. By mapping detection logic to ATT&CK techniques, analysts can better understand what stages of an attack lifecycle are being monitored. According to Al-Sada, Sadighian and Oligeri (2024), the ATT&CK framework has emerged as a de facto knowledge base for developing detection strategies and enhancing visibility into attacker behaviour.

## 2.3 SIEM Architectures

Security Information and Event Management (SIEM) systems are a backbone of modern security ops, aggregating and analysing logs from multiple sources. Traditional SIEM architectures collect log data through a centralized mechanism in which agents at endpoints send event data to a central monitoring server. >>> this logs then get parsed and standardized so that different log file formats can be understood in a unified manner. Once you normalize the data, correlation rules can be used to detect suspicious behavior across various systems. As an example, you might see repeated authentication failures followed by various privilege

escalation events that could indicate a compromised account. SIEM platforms also provide dashboards, alerts, and long-term log retention for forensic investigations. As Scarfone and Mell (2007) point out, log analysis performed centrally enhances an organisation's capability to detect intrusions and respond effectively to incidents.

## 2.4 Automated Response and SOAR Concepts

Security Orchestration, Automation and Response (SOAR) technologies expand on traditional monitoring capabilities by allowing automatic actions to be performed in the event that a threat is identified. The automated response mechanisms use predefined playbooks which outline the actions that need to be taken when certain security alerts are triggered. Such actions could include isolating affected systems, disabling suspicious accounts, or alerting security personnel. Nevertheless, automated response systems should be implemented cautiously to minimize inadvertent effects on these legitimate activities. Many organisations use gating mechanisms or human-in-the-loop processes to approve high-risk remediation actions before they are performed. This technique allows to avoid false-positives leading to some auto-response action. There's a strong push, as security environments become increasingly complex, that automation is the key to speeding up incident response times and reducing stress on already strained security teams (Singer and Friedman, 2013).

## 2.5 Reference Frameworks

Numerous well-known cyber security frameworks can be employed to design appropriate security monitoring and responsive architectures. NIST Special Publication 800-53 describes security controls for information systems covering logging, monitoring, and incident response capabilities. In a similar vein, NIST SP 800-61 offers computer security incident handling process guidance. Centralized log collection, privileged account monitoring, and continuous security monitoring are always stressed in CIS Critical Security Controls. CIS Benchmarks offer configuration guidelines designed to enhance operating system and application security for organisations. ISO 27001 also provides an international standard for information security management systems based on risk and continuous improvement of the security posture. Mapping security monitoring architectures to these frameworks can help ensure that technical design implements accepted industry best practice.

## 2.6 Literature Synthesis

Integrating endpoint telemetry with centralized monitoring and automated response mechanisms is critical for effective detection and containment of security threats, as presented in the literature reviewed. When it comes to endpoint security principles, expected practices include strong logging and controlled system privileges, whilst detection engineering techniques come into play in order to make a determination on attacker behaviour using telemetry data, typically crafted as MITRE ATT&CK frameworks. Security Information and Event Management (SIEM) architectures allow for centralized analysis and correlation of logs across systems, which helps to increase visibility into potential security incidents. Moreover, automation and SOAR ideas allow for quicker response times by providing predefined remediation actions. These ideas greatly influenced design decisions made in this project, which combines endpoint telemetry collection, SIEM-based detection with Wazuh and automated containment through a containerised remediation platform.

# 3. Scenario and Threat Model

## 3.1 Company overview

The organisation simulated, the Larkspur Retail Group works in the transportation and warehousing industry. The company uses the services of endpoint systems to control operational processes, inventory tracking, and access to corporate resources by the company employees.

The simulated infrastructure used in this project consisted of three primary components:

• A Windows workstation representing a typical employee endpoint

• A Linux server hosting the Wazuh SIEM platform

• A containerised automation environment responsible for analysing alerts and executing remediation commands

## 3.2 Incident narrative

The simulated attack scenario entailed creation of a new user account followed by the delegation of administrative privileges (al., 2024). This category of action is normally associated with insider threats or intruders who have obtained lawful credentials.

The assumption was that the assailant model already had access to a standard user account and attempted to acquire further privileges to obtain full administrative privileges of the machine.

## 3.3 Assumed adversary

| **Category** | **Description** |
|---|---|
| Adversary Type | Insider threat or external attacker with compromised user credentials |
| Initial Access Capability | Ability to log in using a standard user account on the Windows endpoint |
| Technical Skills | Moderate technical capability, familiar with command-line tools and privilege escalation techniques |
| Objectives | Escalate privileges, gain administrative access, and maintain persistence on the endpoint |
| Tools and Techniques | Creation of new user accounts, privilege escalation commands, modification of administrative group membership |
| Likely TTPs | Account manipulation, privilege escalation, credential abuse, and persistence mechanisms |
| MITRE ATT&CK Techniques | Valid Accounts (T1078), Account Manipulation (T1098), Privilege Escalation (T1068) |
| Operational Constraints | Activities are limited to the simulated lab environment and do not include large-scale network exploitation |
| Detection Risk | High likelihood of generating detectable logs due to monitored authentication and administrative actions |

## 3.4 Assets, trust boundaries, and data classification

| **Asset / Component** | **Description** | **Importance Level** | **Trust Boundary** | **Telemetry Source** |
|---|---|---|---|---|
| Windows Endpoint | Employee workstation used for normal operations | High | Internal user network | Windows Event Logs, authentication logs |

| | | | | |
|---|---|---|---|---|
| Linux Monitoring Server | Hosts Wazuh SIEM platform for log collection and analysis | Critical | Security monitoring zone | Wazuh logs, system logs |
| Wazuh Agent | Endpoint telemetry collection component | High | Endpoint to SIEM boundary | Security events and system logs |
| Docker Automation Stack | Executes automated containment and remediation actions | High | Security automation boundary | Container logs and remediation logs |
| Administrative Accounts | Privileged user accounts controlling system configuration | Critical | Privilege boundary | Authentication and privilege escalation logs |

## 3.5 Attack paths and hypotheses

| **Attack Stage** | **Hypothesised Activity** | **Description** | **Expected Logs / Evidence** |
|---|---|---|---|
| Initial Access | Compromised user credentials | Attacker logs into Windows endpoint using valid credentials | Login events in Windows security logs |
| Privilege Escalation | Creation of new user account | Attacker creates a new user account on the system | Account creation events |
| Privilege Abuse | Adding user to administrators group | Newly created account is assigned administrative privileges | Group membership modification logs |
| Persistence | Maintaining administrative access | Attacker maintains access through privileged account | Repeated authentication activity |

| | | | |
|---|---|---|---|
| Detection Trigger | SIEM rule activation | Detection rules identify suspicious administrative behaviour | Wazuh alert events |
| Containment | Automated remediation | Docker-based automation restricts privileges of suspicious account | Container logs and remediation logs |

# 4. Methodology

This section explains the experimental setup used to implement the endpoint monitoring and automated containment architecture. The methodology includes the configuration of a controlled virtual laboratory environment, deployment of telemetry collection mechanisms, integration with the SIEM monitoring platform, and the design of automated detection and remediation processes. The approach allows the safe simulation of suspicious administrative behaviour while ensuring that all activities remain isolated from real organisational systems.

## 4.1 Lab Environment Build

A virtualised laboratory environment was created to simulate a small enterprise infrastructure. The environment was built using a hypervisor to deploy separate virtual machines for endpoint monitoring and SIEM analysis.

| **Component** | **Specification** | **Purpose** |
|---|---|---|
| Hypervisor | Oracle VirtualBox | Used to create isolated virtual machines |
| Windows VM | Windows 10 workstation | Simulated employee endpoint generating security events |
| Linux VM | Ubuntu Server | Hosted the Wazuh SIEM monitoring platform |
| CPU Allocation | 2–4 virtual cores per VM | Ensured stable operation of monitoring services |
| Memory | 4–8 GB RAM | Supported log processing and container execution |
| Storage | 40–60 GB virtual disk | Stored logs and SIEM data |

**Network Configuration**

| Network Component | Configuration |
|---|---|
| Network type | Internal virtual network |
| IP addressing | Static IP assignment for monitoring server |
| NAT | Enabled for external package downloads |
| Port forwarding | Used for SIEM dashboard access |
| Time synchronization | System clocks synchronized to ensure accurate log timestamps |

The internal network design ensured that testing activities remained fully isolated while allowing controlled communication between the endpoint and the monitoring server.

## 4.2 Vulnerable Configuration Choices

To generate realistic security events, several intentional weaknesses were introduced within the simulated environment. These vulnerabilities were designed to replicate conditions that attackers commonly exploit in real-world systems.

| Vulnerability | Description | Purpose |
|---|---|---|
| Weak administrative password | Simplified credentials on the endpoint | Simulate credential compromise |
| Privilege escalation permissions | Ability to modify administrator group membership | Generate privilege escalation events |
| Limited access controls | Default user account permissions | Simulate insider threat behaviour |
| Reduced endpoint restrictions | Relaxed security policies | Allow suspicious activity generation |
| Minimal service restrictions | Standard services enabled | Generate system activity logs |

## 4.3 Telemetry Sources

Security telemetry was collected from both Windows and Linux systems to support event correlation and detection.

| Telemetry Source | Platform | Data Collected |
|---|---|---|
| Windows Event Logs | Windows endpoint | Authentication events, account changes |
| Security Logs | Windows endpoint | Privilege escalation events |
| System Logs | Linux monitoring server | Authentication and administrative activity |
| Wazuh Agent Logs | Endpoint agent | Security alerts and monitoring data |
| Authentication Logs | Both systems | Login attempts and account usage |
| Process Events | Endpoint system | Suspicious process activity |

## 4.4 SIEM Integration

The monitoring architecture relied on the Wazuh SIEM platform to collect, analyse, and visualise endpoint telemetry.

| Component | Function |
|---|---|
| Wazuh Agent | Installed on the Windows endpoint to collect system logs |
| Wazuh Manager | Central server responsible for log processing |
| Ingestion Pipeline | Transfers endpoint logs to SIEM platform |
| Parsing Rules | Convert raw logs into structured security events |
| Alert Engine | Generates alerts when suspicious activity occurs |
| Log Retention | Stores historical logs for analysis |

## 4.5 Detection Engineering Approach

Detection engineering focused on identifying suspicious administrative activities using rule-based monitoring.

| Detection Use Case | Log Source | MITRE ATT&CK Technique | Trigger Condition |
|---|---|---|---|

| | | | |
|---|---|---|---|
| Account creation monitoring | Windows logs | T1078 – Valid Accounts | New user account created |
| Privilege escalation detection | Security logs | T1068 – Privilege Escalation | User added to administrator group |
| Authentication anomalies | Authentication logs | T1110 – Brute Force | Multiple login attempts |
| Suspicious administrative actions | System logs | T1098 – Account Manipulation | Administrative group modification |

## 4.6 Automation and Remediation Design

To improve response speed, an automated containment mechanism was implemented using containerised services.

| **Component** | **Function** |
|---|---|
| Python Analysis Module | Evaluates incoming SIEM alerts |
| Decision Engine | Determines alert severity |
| Remediation Script | Executes containment actions |
| Docker Container | Runs automated response services |

## 4.7 Validation and Evaluation

To verify system functionality, a controlled attack scenario was simulated.

| **Test Case** | **Expected Outcome** |
|---|---|
| Creation of new user account | SIEM generates account creation alert |
| Adding account to administrator group | Privilege escalation alert triggered |
| Authentication attempts | Authentication log events recorded |
| Automated remediation | Docker automation executes containment script |

## 4.8 Ethical and Safety Considerations

All testing activities were conducted within an isolated laboratory environment to ensure safe experimentation.

| Ethical Consideration | Implementation |
|---|---|
| Lab isolation | Virtual network separated from external systems |
| Controlled testing | Only simulated attacks were executed |
| No real malware | Testing used benign command simulations |
| Data protection | No real organisational data used |
| Responsible experimentation | Activities limited to academic research purposes |

# 5. System Architecture and Implementation

This section describes the architecture and technical implementation of the endpoint monitoring and automated containment system developed in the simulated laboratory environment (Basta, et al., 2025). The architecture integrates endpoint telemetry collection, SIEM-based monitoring using Wazuh, and an automated remediation layer implemented through a containerised automation stack.

## 5.1 High Level Architecture

The system architecture consists of four main components: monitored endpoints, telemetry collection agents, a centralized SIEM platform, and an automated response layer implemented using Docker containers.

| Component | Description | Role |
|---|---|---|
| Windows Endpoint | Employee workstation generating system activity | Produces security telemetry |
| Linux Monitoring Server | Hosts Wazuh manager and monitoring services | Central log processing |
| Wazuh Agent | Installed on endpoint system | Collects and forwards logs |
| Wazuh Manager | SIEM engine for correlation and alert generation | Security monitoring |

| | | |
|---|---|---|
| Docker AI Automation Stack | Containerised remediation engine | Automated response actions |

**Data Flow**

**Step Description**

1 Endpoint generates authentication and system events

2 Wazuh agent collects logs from the endpoint

3 Logs transmitted to Wazuh manager

4 SIEM engine processes and correlates events

5 Alerts generated when suspicious activity detected

6 Docker automation stack evaluates alerts

7 Automated remediation actions executed

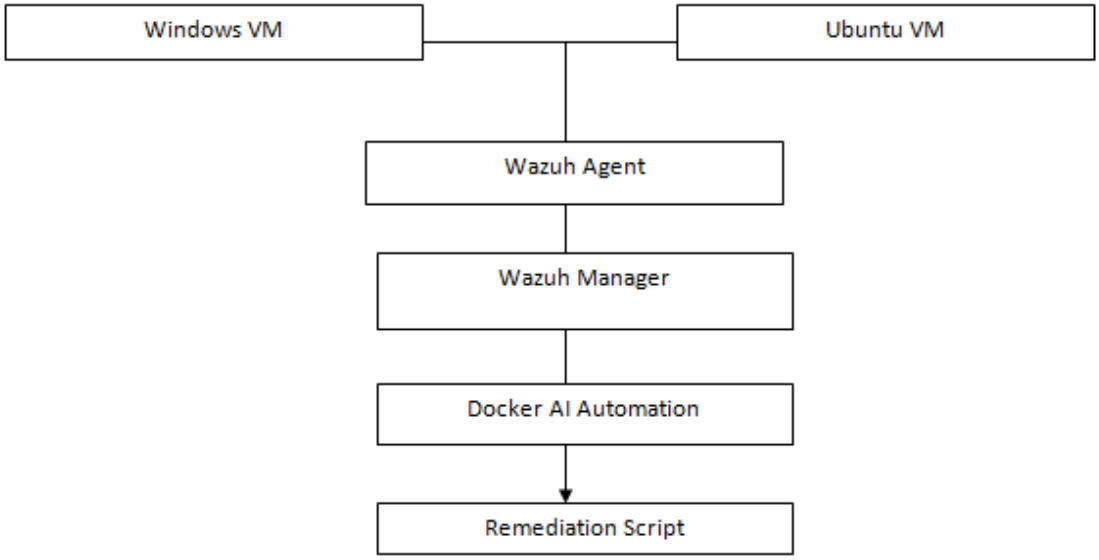


**Figure 1: Architecture of the stack**

## 5.2 Endpoint Configurations

Endpoint systems were configured to generate detailed telemetry that could support detection of suspicious administrative activity.

**Windows Endpoint Configuration**

| Configuration | Description |
|---|---|
| Security event logging | Captures authentication and account management events |
| Windows audit policy | Logs privilege escalation and user account changes |
| PowerShell logging | Captures execution of administrative commands |
| Local administrator monitoring | Detects group membership modifications |
| Defender configuration | Default endpoint security monitoring enabled |

**Linux Monitoring Server Configuration**

| Configuration | Description |
|---|---|
| System logging | Authentication and administrative activity monitoring |
| Journald logging | Captures system service events |
| SSH configuration | Restricted remote access |
| Sudo monitoring | Detects privilege escalation attempts |
| Log collection | Central storage of endpoint telemetry |

These configurations ensured that key events such as account creation, login attempts, and privilege escalation actions were captured by the monitoring infrastructure.

## 5.3 SIEM Configuration

The Wazuh SIEM platform was configured to collect logs from monitored endpoints and perform event correlation to detect suspicious behaviour.

**Ingestion and Indexing**

| Component | Function |
|---|---|
| Wazuh Agent | Sends logs from endpoint systems |
| Log Parser | Converts raw logs into structured fields |

| | |
|---|---|
| Event Indexing | Stores logs in searchable format |
| Timestamp normalization | Ensures consistent log timing |
| Host identifiers | Associates logs with monitored hosts |

**Dashboards and Visualizations**

The SIEM platform provides visual dashboards for monitoring authentication activity and suspicious behaviour.

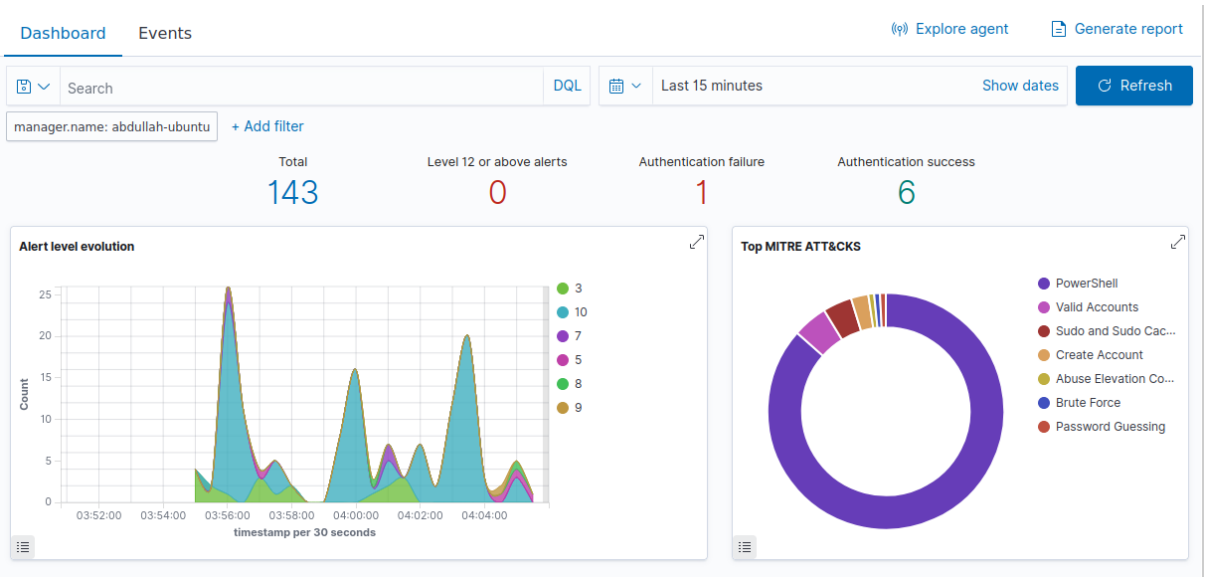


**Figure 2 – Wazuh monitoring dashboard displaying aggregated security alerts**

This figure shows the dashboard containing alert counts, authentication activity, and MITRE ATT&CK classifications.

| Time | Agent | Agent name | Technique(s) | Tactic(s) | Description | Level | Rule ID |
|---|---|---|---|---|---|---|---|
| Mar 9, 2026 @ 04:07:08.130 | 008 | abdullah-Win | T1059.001 | Execution | Windows account created | 10 | 777111 |
| Mar 9, 2026 @ 04:05:31.712 | 000 | abdullah-ubuntu | | | Crontab entry changed. | 5 | 2832 |
| Mar 9, 2026 @ 04:05:25.704 | 000 | abdullah-ubuntu | T1136 | Persistence | Multiple crontabs edits detected | 10 | 777555 |
| Mar 9, 2026 @ 04:05:23.703 | 000 | abdullah-ubuntu | | | Crontab entry changed. | 5 | 2832 |
| Mar 9, 2026 @ 04:05:15.815 | 000 | abdullah-ubuntu | T1136 | Persistence | Multiple crontabs edits detected | 10 | 777555 |
| Mar 9, 2026 @ 04:05:09.688 | 000 | abdullah-ubuntu | T1136 | Persistence | Multiple crontabs edits detected | 10 | 777555 |
| Mar 9, 2026 @ 04:05:01.016 | 008 | abdullah-Win | | | Custom Detection: Windows Software Protection Service Activity Detected | 8 | 777333 |
| Mar 9, 2026 @ 04:04:55.675 | 000 | abdullah-ubuntu | T1548 | Privilege Escalation, Defense Evasion | Failed attempt to switch user | 9 | 777444 |

Security Alerts

| Time | Agent | Agent name | Technique(s) | Tactic(s) | Description | Level | Rule ID |
|---|---|---|---|---|---|---|---|
| Mar 9, 2026 @ 04:04:29.317 | 008 | abdullah-Win | T1059.001 | Execution | Windows account created | 10 | 777111 |
| Mar 9, 2026 @ 04:03:56.994 | 008 | abdullah-Win | T1110 | Credential Access | Multiple failed login attempts | 10 | 777222 |

**Figure 3 – Security event logs showing triggered detection rules**

## 5.4 AI Security Stack in Docker

To improve response capabilities, an automated remediation mechanism was implemented using Docker containers (J. Myers, 2011).

**Component Description**

| Component | Function |
|---|---|
| Python Analysis Module | Evaluates alerts received from SIEM |
| Decision Engine | Determines severity level of alerts |
| Remediation Script | Executes containment actions |
| Docker Container | Hosts automation services |

The automation system analyses SIEM alerts and triggers predefined remediation actions when critical events are detected.

**Docker Compose Overview**

| Component | Description |
|---|---|
| Services | Python automation service |
| Volumes | Store logs and remediation records |
| Network | Internal container communication |
| Resource limits | Restrict container CPU and memory usage |
| Secrets handling | Secure storage of configuration values |

**Security Controls**

| Control | Implementation |
|---|---|
| Least privilege containers | Containers run with minimal permissions |

| | |
|---|---|
| Restricted file access | Read-only filesystem where possible |
| No privileged containers | Prevent full host access |
| Logging enabled | All remediation actions logged |

```
ubuntu@abdullah-ubuntu:~/PartD_Project/automation-demo$ ./demo.sh
Starting AI Automation Stack...
Removing network automation-demo_default
Creating network "automation-demo_default" with the default driver
Building automation
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
            Install the buildx component to build images with BuildKit:
            https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  4.608kB
Step 1/6 : FROM python:3.9-slim
 ---> 085da638e1b8
Step 2/6 : WORKDIR /app
 ---> Using cache
 ---> 1b307e6c05a3
Step 3/6 : RUN pip install --no-cache-dir flask
 ---> Using cache
 ---> 28e89e4f0199
Step 4/6 : COPY . .
 ---> Using cache
 ---> f8fb47d17418
Step 5/6 : EXPOSE 5000
 ---> Using cache
 ---> 4278d8b8561a
Step 6/6 : CMD ["python", "app.py"]
 ---> Using cache
 ---> f8890b1227a3
Successfully built f8890b1227a3
Successfully tagged automation-demo_automation:latest
Creating automation ... done
Waiting for container to start...
-------------------------------------
Simulating Wazuh Custom Rule 110001
-------------------------------------
{"remediation":"User disabled","rule_id":"777111"}


-------------------------------------
Simulating Multiple Failed Logins
-------------------------------------
```

```
 ---> Using cache
 ---> 1b307e6c05a3
Step 3/6 : RUN pip install --no-cache-dir flask
 ---> Using cache
 ---> 28e89e4f0199
Step 4/6 : COPY . .
 ---> Using cache
 ---> f8fb47d17418
Step 5/6 : EXPOSE 5000
 ---> Using cache
 ---> 4278d8b8561a
Step 6/6 : CMD ["python", "app.py"]
 ---> Using cache
 ---> f8890b1227a3
Successfully built f8890b1227a3
Successfully tagged automation-demo_automation:latest
Creating automation ... done
Waiting for container to start...
-------------------------------------
Simulating Wazuh Custom Rule 110001
-------------------------------------
{"remediation":"User disabled","rule_id":"777111"}


-------------------------------------
Simulating Multiple Failed Logins
-------------------------------------
{"remediation":"IP blocked","rule_id":"777222"}


-------------------------------------
Check remediation log:
-------------------------------------
2026-03-08 22:51:53.689209 - User disabled.
2026-03-08 22:51:55.705156 - IP blocked.

-----------------------------------------
Demo Completed Successfully
-----------------------------------------
ubuntu@abdullah-ubuntu:~/PartD_Project/automation-demo$
```

**Figure 4 – Container log output showing automated remediation execution**

# 6. Results

## 6.1 Evidence of Compromise in the Lab

The simulated attack involved the creation of a new user account followed by multiple user logins with wrong passwords (Amjad, 2024). These actions generated security events that were captured by the SIEM monitoring system.

The SIEM platform successfully detected suspicious behaviour, including account creation events and authentication anomalies.

## 6.2 Detection Outcomes

| **Use Case** | **Log Source** | **ATT&CK Technique** | **Trigger Condition** | **Severity** | **Result** |
|---|---|---|---|---|---|
| Account Creation | Windows Logs | T1059.001 | New user account created | High | Alert generated |
| Failed Logins | Authentication Logs | T1110 | Multiple login attempts | High | Alert generated |
| Windows Software Protection Service | Windows Logs | | Windows Software Protection Service detected | Medium | Alert generated |
| Privilege Escalation | Linux Logs | T1548 | Attempt to switch user | Medium | Alert generated |
| Cron Modification | System Logs | T1136 | Multiple cron edits | High | Alert generated |

## 6.3 Automated Remediation Outcomes

| **Trigger Event** | **Remediation Action** | **Verification** |
|---|---|---|
| Windows account creation | User account disabled | Container logs confirm action |
| Multiple failed logins | Source IP blocked | Firewall rules updated |

## 6.4 Performance and Reliability

| Metric | Observation |
|---|---|
| Log ingestion speed | Real-time log transmission observed |
| Detection latency | Alerts generated within seconds |
| Container stability | No crashes during automation tests |
| Log completeness | All simulated events captured |

## 6.5 Limitations

| Limitation | Description |
|---|---|
| Limited endpoints | Only one monitored endpoint |
| Simplified attack scenario | Real attackers may use more complex techniques |
| Reduced telemetry | No enterprise EDR integration |
| Small-scale SIEM deployment | Production systems require larger infrastructure |

# 7. Discussion

The SIEM monitoring architecture adopted in this project proved usefulness of integrating SIEM monitoring and automated response mechanisms.

The capability to identify the suspicious behaviour as soon as it had taken place was one of the biggest benefits of the system. The generation of administrative privilege changes created high severity alerts in the SIEM platform, allowing the detection of possible security incidents quickly.

The automated remediation further enhanced response time through the automated nature of the remediation component. Containment measures were automatically implemented after the alert was declared critical.

Despite these benefits, there are a number of weaknesses that should be considered. Automated remediation must be applied carefully so that normal system operations are not interrupted. Enterprise world security teams are more likely to incorporate approval workflow before high-impact remediation can be effected.

Additionally, the virtual environment used in this project was limited in endpoints. The extent of deployment by an enterprise would require more complex monitoring infrastructure and more complex detection logic.

In a number of ways, this security posture of the simulated organisation could be enhanced.

Firstly, it would enable more effective logging mechanisms, such as Microsoft Sysmon, which would provide more useful telemetry data and could be better able to detect it.

Secondly, administrative account multi-factor authentication would significantly reduce exposure to credential abuse (D. R. Garcia, 2024).

Thirdly, administrative privileges must be applied in a more strict manner to minimize the amount of users that have privileged access credentials at organizations.

Finally, detection rules should be repeatedly checked and updated in regard to evolving attack methods.

# 8. Conclusion

This project demonstrated how to design and deploy an endpoint monitoring and automated containment architecture on a simulated enterprise environment.

Under the automation environment that saw the deployment of the Wazuh SIEM platform with a containerised automation system, the system could detect any suspicious administrative activities and automatically instigate the process of containment.

According to the experiment, organisational cybersecurity capacity can be expanded through the incorporation of monitoring, detection engineering, and automated response technologies.

The system might be expanded later on to accommodate additional sources of telemetry, behavioural analytics and better integration of threat intelligence.

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# Appendix

## Appendix A – System Architecture Overview

| Component | Description | Role in Architecture |
|---|---|---|
| Windows Endpoint | Virtual machine simulating an employee workstation | Generates authentication and administrative activity logs |
| Linux Monitoring Server | Ubuntu server hosting the Wazuh SIEM platform | Centralised log processing and monitoring |
| Wazuh Agent | Endpoint telemetry collector installed on Windows VM | Sends security logs to SIEM |
| Wazuh Manager | Core SIEM component running on the Linux server | Processes, correlates, and generates alerts |
| Docker Automation Stack | Containerised automation environment | Executes remediation actions based on SIEM alerts |
| Remediation Scripts | Python scripts executed by Docker containers | Perform automated containment actions |

## Appendix B – Virtual Lab Network Configuration

| Parameter | Configuration |
|---|---|
| Hypervisor | Oracle VirtualBox |
| Network Type | Internal virtual network |
| Windows VM IP Address | 192.168.56.10 |
| Linux SIEM Server IP | 192.168.56.20 |
| Communication Protocol | TCP |
| SIEM Port Access | Local dashboard access enabled |
| Time Synchronisation | Enabled between VMs |

## Appendix C – Detection Rules Implemented

| Detection Rule | Log Source | ATT&CK Technique | Trigger Condition | Severity |
|---|---|---|---|---|
| Windows Account Creation | Windows Security Logs | T1059.001 | Creation of new user account | High |

| | | | | |
|---|---|---|---|---|
| Multiple Failed Login Attempts | Authentication Logs | T1110 | Multiple authentication failures | High |
| Privilege Escalation Attempt | Linux System Logs | T1548 | Attempt to switch user privileges | Medium |
| Cron Modification Detection | Linux Logs | T1136 | Multiple crontab modifications detected | High |

## Appendix D – Automated Remediation Actions

| **Trigger Event** | **Detection Rule ID** | **Automated Response** | **Verification Method** |
|---|---|---|---|
| New Windows account created | 777111 | Disable suspicious user account | Container remediation log |
| Multiple failed logins | 777222 | Block source IP address | Firewall rule confirmation |
| Privilege escalation attempt | 777444 | Restrict administrative privileges | SIEM alert confirmation |

## Appendix E – Docker Automation Stack Components

| **Component** | **Technology Used** | **Function** |
|---|---|---|
| Automation Container | Docker | Hosts remediation services |
| Python Analysis Script | Python | Evaluates SIEM alerts |
| Decision Engine | Rule-based logic | Determines severity of alerts |
| Remediation Script | Bash/Python | Executes containment actions |
| Log Storage | Docker volumes | Stores remediation logs |

### Appendix F – Test Case Scenarios

| **Test Case** | **Simulated Action** | **Expected Detection** |
|---|---|---|
| User Account Creation | New Windows user account created | SIEM alert triggered |

| | | |
|---|---|---|
| Failed Authentication Attempts | Multiple login failures | Brute-force detection alert |
| Remediation Test | Automation script executed | Account disabled / IP blocked |

**Appendix G – Security Monitoring Metrics**

| **Metric** | **Observation** |
|---|---|
| Detection Time | Alerts generated within seconds |
| Log Ingestion Speed | Near real-time |
| Container Stability | No failures observed |
| Alert Accuracy | All simulated attacks detected |